PHYS 5041: Course Content

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Differential Equations I, `diffeq-I.pdf`

- Terminology: independent / dependent variables, linear / non-linear, order, homogeneous / inhomogeneous.
- Solving first-order ODEs: separable equations, linear equations (integrating factor), exact differentials, change of variables.
- Solving second-order ODEs: homogeneous equations with constant coefficients, Euler's Equation, inhomogeneous equations, homogeneous boundary conditions.
- Green Function.
- Hypergeometric Differential Equation, ${}_2F_1(a, b, c; x)$.
- Dirac Delta Function.
- Differentiating definite integrals.

Series, `series.pdf`

- Geometric series, binomial series.
- Convergence of series: preliminary test, comparison test, ratio test, integral test, alternating series, absolute / conditional convergence.
- Power series, Taylor series, L'Hôpital's rule.

Differential Equations II, `diffeq-II.pdf`

- Wronskian and its use for finding the second solution.
- Solution of second-order ODEs via generalized power series (Method of Frobenius), singular points, Fuch's Theorem, indicial equation, Frobenius method solution of the Hypergeometric Differential Equation.
- Green Function.

Orthogonal Functions, `orthogonal-functions.pdf`

- Orthogonality, completeness.
- Sturm-Liouville ODE, boundary conditions.
- Inner product.
- Expanding the Green Function in orthogonal functions.

Partial Differential Equations, `pdes.pdf`

- Identify named equations.
- Separation of variables technique.
- Matching boundary conditions with the help of orthogonal functions.

Fourier Series and Transforms, `fourier.pdf`

- Fourier series, exponential and trigonometric forms. Even and Odd functions.
- Parseval's theorem for Fourier series.
- Fourier transform, Parseval's theorem for Fourier transforms, convolution.
- Discrete Fourier transforms, convolution.

Legendre Polynomials and Related Topics, `legendre.pdf`

- Separation of the Laplacian in spherical coordinates.
- Legendre differential equation, Frobenius solution.
- Orthogonality and completeness of Legendre polynomials.
- Solving the Laplace equation for problems with azimuthal symmetry.
- Generating function recurrence formulas, Rodriges' formula, second solution.
- Associated Legendre functions and the $Y_\ell^m(\theta, \phi)$.
- General solution to the Laplace equation when $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

Gamma function, `gamma.pdf`

- Definition and basic properties.
- Stirling's formula.

Bessel functions, `bessel.pdf`

- Separation of the Laplacian in cylindrical coordinates.
- The Bessel ODE and its solutions $J_\nu(x)$ and $N_\nu(x)$.
- Orthogonality and completeness of the $J_\nu(x)$.
- Modified Bessel functions $I_\nu(x)$ and $K_\nu(x)$.
- Limiting forms for large and small x for the functions $J_\nu(x)$, $N_\nu(x)$, $I_\nu(x)$, and $K_\nu(x)$.
- Boundary value PDEs in cylindrical coordinates.
- Spherical Bessel functions.
- Helmholtz equation in spherical coordinates.

Wave equation, `wave-equation.pdf`

- Separation of variables.
- Solutions generally involve sinusoidal time dependence and then the Helmholtz equation.

Complex variable theory, `complex-variables.pdf`

- Analytic functions, Cauchy-Riemann equations.
- Evaluating powers, logarithms, and exponentials of complex arguments.
- Contour integrals, Cauchy's theorem, Laurent series, Residue theorem.
- Finding residues, evaluating definite integrals using contour integration.
- Branch points and branch cuts.